

Piyush Hingu

☎ +91 9408926001 | ✉ CS22M107@iittp.ac.in | 🔗 [linkedin.com/in/piyush-hingu](https://www.linkedin.com/in/piyush-hingu) | 🐙 github.com/piyushhingu

PROFESSIONAL SUMMARY

Software Engineer experienced in developing C++/C applications for hardware measurement systems by accessing NVIDIA and vendor-provided APIs. Skilled in creating Python-based visualization and testing tools that analyze data generated from these APIs. Complementary expertise includes CUDA and PTX-based LLM optimization and benchmarking, focusing on kernel integration and performance profiling rather than kernel implementation. Collaborative experience with FPGA and hardware teams for real-time firmware-driver interfacing and data synchronization.

Core Skills: C++ / C API integration • Hardware measurement and waveform analysis • NVIDIA API data access • Python visualization and automation • CUDA GEMM and LLM kernel benchmarking • Real-time performance profiling • FPGA collaboration

TECHNICAL SKILLS

Programming Languages: C++, C, Python, CUDA C++, PTX Assembly, C

Frameworks/Tools: Visual Studio, .NET, NVIDIA Nsight Systems, Jenkins, CMake, Docker

Measurement & Analysis: Power and waveform measurement, harmonic and switching behavior evaluation, compliance testing

Visualization: Python (Matplotlib, NumPy), REST APIs, real-time data visualization

ML & Kernel Optimization: CUDA GEMM, tensor cores (WMMA), kernel benchmarking and profiling, LLM inference stack integration

PROFESSIONAL EXPERIENCE

Qualcomm Technologies

Jan 2026 – Present

System Integration & Testing Engineer(SIT)

- Performed system integration and validation of large language model (LLM) inference pipelines on Qualcomm platforms, focusing on stability, scalability, and performance.
- Tested and benchmarked diverse model architectures including Mixture-of-Experts (MoE) and audio-based models using inference frameworks such as vLLM and Efficient Transformers.
- Evaluated advanced serving features including disaggregated serving and compute context length (CCL) under production-like workloads.
- Collected and analyzed key performance metrics including Time-To-First-Token (TTFT), Time-Per-Output-Token (TPOT), throughput, decode latency, and token generation efficiency.
- Designed and executed automated test pipelines across varying batch sizes, sequence lengths, and hardware configurations.
- Collaborated with model, runtime, and hardware teams to debug integration issues and improve end-to-end inference performance.

Zentreelabs Pvt. Ltd.

June 2024 – Dec 2025

R & D Engineer

- Developed C++/C applications interfacing with NVIDIA and hardware vendor APIs to access real-time measurement and performance data from GPU and FPGA systems.
- Built Python visualization and test automation tools for waveform evaluation, power analysis, and compliance testing.
- Optimized LLM inference pipelines by integrating and benchmarking CUDA GEMM kernels using NVIDIA APIs, focusing on profiling and throughput analysis.
- Automated benchmarking using Jenkins and Nsight Compute to validate latency, throughput, and utilization across LLM backends (vLLM, CTranslate2, ONNX/TensorRT).
- Collaborated with FPGA and hardware engineers to improve firmware-driver communication and data synchronization.

EDUCATION

Indian Institute of Technology Tirupati

M.Tech in Computer Science and Engineering

Mar 2022 – Jun 2024

CGPA: 8.13/10

Gujarat University

B.Tech in Computer Science and Engineering

Apr 2017 – Jun 2021

CGPA: 8.57/10

KEY PROJECTS

Hardware Measurement API Access and Visualization (C++/Python)

2025

- Used NVIDIA and hardware vendor APIs to obtain real-time measurement data including power waveforms and harmonic distortion metrics.
- Developed Python visualization tools to analyze FPGA- and GPU-collected hardware data with real-time plotting for validation and compliance.
- Implemented signal processing algorithms (harmonics, FFT) in Python and C++ to process measurement data efficiently.

CUDA GEMM Integration for LLM Inference Benchmarking

2025

- Integrated and benchmarked vendor-provided CUDA GEMM kernels within transformer inference stacks (vLLM, CTranslate2) using NVIDIA APIs.
- Conducted performance profiling with Nsight Compute focusing on memory usage, occupancy, and throughput optimizations.
- Automated benchmarking and validation pipelines with Jenkins CI/CD for scalable LLM model deployment testing.

LLM Serving Stack Optimization

2024

- Implemented dynamic batching and quantization-aware workflows to compress and accelerate large language model inference.
- Containerized inference backends (Docker, Kubernetes) and automated health monitoring and scaling for production readiness.

CERTIFICATIONS

Introduction to Natural Language Processing (IIT Tirupati)